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Key Points:

- The South Qiangtang block was located at ~22°S in the middle Permian
- The width of the Paleo-Tethys exceeded 5,500 km in the middle Permian, when the width of the Neo-Tethys was ~1,000–2,000 km
- The South Qiangtang block drifted rapidly northward with the Cimmerian blocks in early Permian to Late Triassic time

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

X. Cheng and and H. Wu,
chengxin@nwu.edu.cn;
wuhn2506@nwu.edu.cn

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Placing Another Piece of the Tethyan Puzzle: The First Paleozoic Paleomagnetic Data From the South Qiangtang Block and Its Paleogeographic Implications

Bitian Wei[1], Xin Cheng[1], Mathew Domeier[2], Nan Jiang[1,3], Yiying Wu[1], Weijie Zhang[1,4], Ke Wu[1], Baofeng Wang[1], Pengxiang Xu[1], Longyun Xing[1], Dongmeng Zhang[1], Teng Li[1], Xiaohong Deng[1], Feifan Liu[1], Yanan Zhou[1], and Hanning Wu[1]

[1]Department of Geology, State Key Laboratory of Continental Dynamics, Northwest University, Xi'an, China, [2]Centre for Earth Evolution and Dynamics (CEED), University of Oslo, Oslo, Norway, [3]School of Petroleum Engineering and Environmental Engineering, Yan'an University, Yan'an, China, [4]Department of Ocean Science and Engineering, Centre for Marine Magnetism (CM^2), Southern University of Science and Technology, Shenzhen, China

Abstract The origin and drift history of the South Qiangtang block (SQB) is critical to understanding the internal dimensions and kinematics of the late Paleozoic to early Mesozoic Tethyan realm and the subsequent evolution of the Qinghai-Tibet Plateau, but so far no paleomagnetic data constrain the SQB in Paleozoic time. Here we present a paleomagnetic study of the middle Permian volcanic and sedimentary rocks from the SQB. Well-defined high-temperature components of magnetization were isolated from 6 sites of tuffaceous sandstone and 9 sites of basalt, but their directional populations are distinct. On the basis of a conglomerate test and comparisons with younger paleomagnetic data, we argue the basalts preserve a primary magnetization, whereas the sandstones may have been remagnetized during the Triassic. The magnetic directions from the basalt indicate that the SQB was located at a latitude of 22.3 ± 5.8°S during the middle Permian. Together with existing paleomagnetic data from other Tethyan continental blocks, this new constraint reveals that during the middle Permian the Neo-Tethys Ocean spanned ~10–20° of latitude and the Paleo-Tethys Ocean spanned ~50°, but also that the latter was a composite ocean system which included at least two important branches. In further considering the geological records of the SQB, we propose the block rifted from the northern margin of Gondwana during opening of the Neo-Tethys in early Permian time, and then drifted rapidly northward until ultimately colliding with the North Qiangtang block and the southern margin of Laurasia during final closure of the Paleo-Tethys in the Triassic.

Plain Language Summary The formation of the Qinghai-Tibet Plateau involved the opening and closing of multiple oceans (the 'Tethyan' oceans) and the rifting, drifting and collision of multiple continental blocks, but quantitative paleogeographic constraints on the motions of these tectonic 'puzzle pieces' are scarce. Here we present paleomagnetic data from one of the central pieces of this puzzle—the South Qiangtang block—which allow us to determine that it was located at a latitude of ~22°S in middle Permian time, 265 million years ago. This paleolatitude constraint further allows us to estimate the width of the Tethyan oceans on either side of the South Qiangtang block in the middle Permian, providing a first-order snapshot of the evolving paleogeographic puzzle that gave rise to the Qinghai-Tibet Plateau.

1. Introduction

The Qinghai-Tibet Plateau comprises a mosaic of geologically distinct Paleozoic and Mesozoic blocks, including the Qaidam block, Songpan-Ganzi block, Qiangtang block(s), Lhasa block and Himalayan block (Figure 1a). It is generally acknowledged that some of these blocks originated from the northern margin of Gondwana, splitting from it in the Carboniferous or Permian before subsequently drifting northward to collide with Laurasia in the Mesozoic (Metcalfe, 2009, 2013; Stampfli et al., 2013; Zhai et al., 2013). These blocks may have rifted from Gondwana in concert with Iran, Afghanistan and other blocks in the west (Sengör et al., 1988) and Sibumasu in the east (Hennig et al., 2009; Peng et al., 2013), together forming an elongate continental ribbon — the 'Cimmerian continent'. This simple tectonic narrative associates the late Paleozoic spalling and northward drift of the Cimmerian continent with the opening and spreading of the Neo-Tethys Ocean, and the later arrival of the Cimmerian continent to the southern margin of Laurasia is taken to mark the closure of the Paleo-Tethys Ocean.

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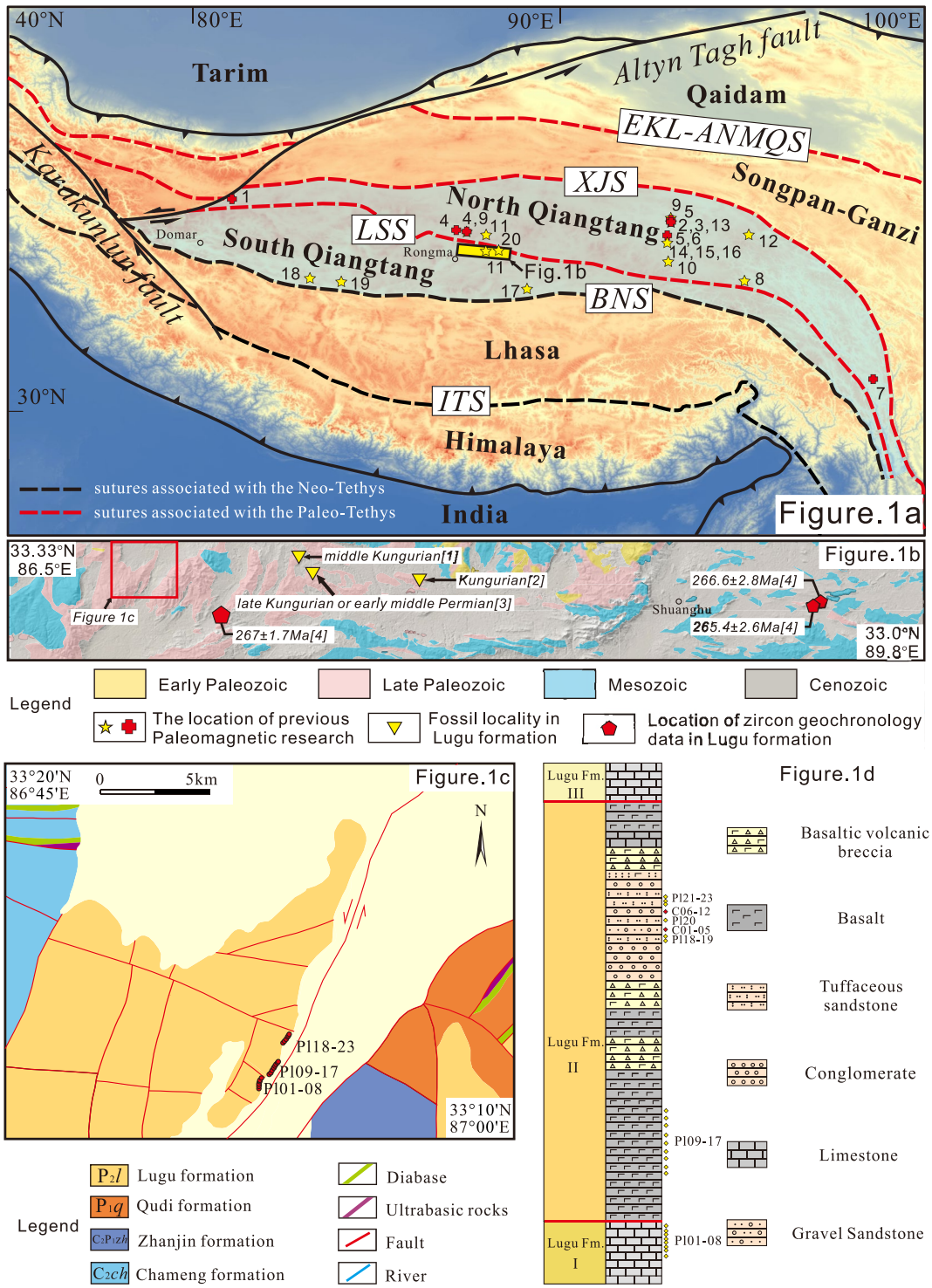


Figure 1.

However, there is a lack of basic quantitative constraints on the time-dependent size of the Neo- and Paleo-Tethys oceans. Furthermore, it is becoming increasingly clear that the specific kinematic evolution of the Tethyan realm was likely more complicated, and involved more than just the northward drift of a single string of continental blocks.

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Of particular importance is the nature and history of the Qiangtang block(s). Simple models of Tethyan evolution have treated the Qiangtang block as a unified block that rifted from northern Gondwana and drifted northward in concert with the rest of the Cimmerian constituents. However, geological investigations have increasingly suggested the Qiangtang block to be a complex of two continental blocks that were independent until early Mesozoic time: a North Qiangtang block (NQB) and a South Qiangtang block (SQB), separated by a Triassic suture (the Longmu Co-Shuanghu Suture Zone; LSS) (e.g., Li, 1987; Liu et al., 2019; Tang & Zhang, 2014; Zhai et al., 2011; Zhang, Cai, et al., 2006; Zhang, Zhang, et al., 2006; Zhang et al., 2016). This potential late Paleozoic independence of the NQB and SQB has important implications for the structure and kinematic evolution of Tethyan realm more broadly, because it implies that the constituents of the Qinghai-Tibet Plateau do not represent a single 'wave' of continental blocks from northern Gondwana. It also challenges the concept of a singular 'Paleo-Tethys Ocean', because both the LSS and the northern boundary of the NQB have been proposed to represent 'Paleo-Tethyan' sutures, in addition to the Eastern Kunlun-Animaqing suture even further north (Figure 1a). Unfortunately, we cannot yet exploit the paleogeographic history of the Qiangtang blocks to better resolve the development of the Tethyan realm because there are not yet sufficient paleomagnetic data constraining their individual drift histories.

A series of paleomagnetic studies have been carried out on the NQB in the last decade, yielding an array of reliable constraints on its drift history from Paleozoic to Mesozoic time (e.g., Cheng, Wu, Diao, et al., 2012; Cheng, Wu, Guo, et al., 2012, Cheng et al., 2013; Guan et al., 2021; Jiang et al., 2022; Ma et al., 2019; Song et al., 2015, 2017; 2020; Yan et al., 2016, Yang et al., 2017; Yu et al., 2022; Zhou et al., 2019). However, paleomagnetic constraints from the SQB are comparably fewer and entirely restricted to Late Triassic and younger time (e.g., Cao et al., 2019, 2020; Chen et al., 2017; Meng et al., 2018; Song et al., 2012), when the Paleo-Tethys Ocean was very nearly or already closed. Thus, there are no quantitative paleogeographic constraints on the SQB in late Paleozoic or earliest Mesozoic time, when that block is thought to have been migrating across the Tethyan realm. Quantitative estimates of the position of the SQB during the late Paleozoic or earliest Mesozoic are therefore critically needed to constrain the dimensions and kinematic evolution of the Tethyan oceans. In this paper, we report the first Paleozoic (middle Permian) paleomagnetic data from the SQB, and apply it to consider both the late Paleozoic-early Mesozoic drift history of the SQB and the relative dimensions of the various branches of the Tethyan oceans.

2. Geological Setting and Paleomagnetic Sampling

Permian strata in the SQB are mainly exposed between the Domar area in the west and Shuanghu County in the east (Zhang, Shi, & Shen, 2013). The sampling area of this study is located approximately 40 km northeast of Rongma Township, Nima County, Tibet, approximately 40 km south of the LSS (Figure 1a). Early Paleozoic, late Paleozoic and Mesozoic strata are exposed in this area (Figure 1b), and Carboniferous and Permian deposits are particularly thick and widely exposed. The Carboniferous series comprises a set of low-grade metamorphic rocks, including the Chameng Formation and the Zhanjin Formation. The Permian series can be divided into two parts according to lithologic changes: a lower assemblage of coarse clastic rocks and sandy limestone is assigned to the early Permian Qudi Formation, whereas the upper part consists mainly of limestone and multi-layered pillow basalts of the middle Permian Lugu Formation. The Carboniferous and Permian strata in the sampling area generally dip to the northeast. However, due to the influence of faults, the units are laterally broken into fault blocks that locally disrupt the regional structure (Figure 1c). Deformation of the Carboniferous and Permian strata is thought to have occurred before the Late Triassic and a north-northeast trending Cenozoic strike-slip fault passes through the study area (Figure 1c) (Zhao et al., 2019).

Figure 1. (a) Regional geological map showing the locations of the major Tibetan blocks, suture zones, and our study area (modified after Yan et al., 2019). The locations of previous paleomagnetic studies are denoted with red crosses (Paleozoic) and yellow stars (Mesozoic), and numbered as follows: 1. Jiang et al., 2022; 2. Yang et al., 2017; 3. Song et al., 2017; 4. Cheng et al., 2012b; 5. Cheng et al., 2013; 6. Ma et al., 2019; 7. Huang et al., 1992; 8. Guan et al., 2021; 9. Zhou et al., 2019; 10. Song et al., 2020; 11. Song et al., 2012; 12. Yu et al., 2022; 13. Song et al., 2015; 14. Lin & Watts, 1988; 15. Cheng, Wu, Diao, et al., 2012; 16. Yan et al., 2016; 17.Cao et al., 2019; 18. Cao et al., 2020; 19. Chen et al., 2017; 20. Meng et al., 2018. Abbreviations: EKL-ANMQS, Eastern Kunlun-Animaqing Suture Zone, XJS: Xijinwulan-Jinshajiang Suture Zone, LSS: Longmu Co-Shuanghu Suture Zone, BNS: Bangong Co-Nujiang Suture Zone, ITS: India-Yarlung Zangbo Suture Zone. (b) Regional geological map (Ryan et al., 2009; Mayigangri, Jiangaidarina, and Tucuo regional geological survey reports) showing the paleomagnetic sampling, fossil and U–Pb dating localities (1: Zhang, Wang, et al., 2012; 2: Zhang et al., 2014; 3: Shen et al., 2016; 4: Li and Wang., 2019). (c) Simplified geologic map of the sampling area (modified after a 1:250,000 Mayigangri regional geological survey). (d) Simplified stratigraphic columns of section PL, showing the locations of the paleomagnetic samples.

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Our paleomagnetic study focuses on the Lugu Formation, which locally is subdivided into three assemblages (Lugu I, II and III; Figure 1d). Lugu I is mainly limestone; Lugu II is dominated by thick pillow basalts with massive basalt and layers of basaltic volcanic breccia, but also includes yellow-gray conglomerate, gravel sandstone, tuffaceous sandstone and a layer of gray brecciated limestone; Lugu III is dominated by thickly-bedded, fine-grained, yellowish-gray limestone (Figure 1d). The Lugu Formation limestones are interpreted to have been deposited on a continental shelf, whereas the volcanics and clastic rocks of Lugu II have been attributed to a continental rifting event (Lai & Qin, 2009; Li & Wang, 2019).

Brachiopods discovered by Shen et al. (2016) in limestones of the Lugu Formation suggest they were deposited in the Kungurian or middle Permian, and Zhang, Shen, et al. (2012), Zhang, Wang, et al. (2012) reported foraminifera *Cancellina primigena* to be dominant in the limestones, further indicating a Kungurian age (Figure 1b). To the west, Yuan et al. (2021) reported a new conodont and fusuline fauna from the lower Lugu Formation limestones, which suggest they were deposited in the late Kungurian. To the east, near Yaqu in Shuanghu County (Figure 1b), Li and Wang (2019) U-Pb dated zircons from basalts and siliciclastic rocks of the Lugu Formation. Their results provided a weighted mean $^{206}Pb/^{238}U$ age of 266.6 ± 2.8 Ma (n = 6, mean square weighted deviation (MSWD) = 0.55) for the crystallization of the basalt, and 19 concordant grains from the siliciclastic rocks yielded an age of 265.4 ± 2.6 Ma (n = 19, MSWD = 2.7). The latter age implies that the siliciclastic rocks were derived from contemporaneous volcanic rocks and rapidly deposited. Another U-Pb date from the Lugu Formation was determined from a basalt in well QZ-5 near Jiaomuchaka, which was dated to 267 ± 1.7 Ma (n = 22, MSWD = 0.18) (Li & Wang, 2019). Based on the above fossil and radiometric evidence, we assign the Lugu I limestones to the Kungurian-Roadian (284-269 Ma), and the basalt and sandstone of Lugu II to the Roadian-Capitanian (270-260 Ma).

We collected paleomagnetic samples from both the lower assemblage (Lugu I) and the middle assemblage (Lugu II) at a section (33.2°N, 86.9°E) hereafter called 'PL'. 8 sites (84 samples) were collected from Lugu I limestones, 9 sites (96 samples) were collected from Lugu II pillow basalts (each site collected from a different flow), 6 sites (73 samples) were collected from Lugu II tuffaceous sandstone, and 12 samples for a conglomerate test were collected from cobbles of basalt in a conglomerate interbedded within the Lugu II tuffaceous sandstones (Figure 1d).

Paleomagnetic samples were collected with use of a gasoline-powered drill, and were oriented in situ using magnetic and solar compasses. A handheld GPS unit was used to determine the geographic coordinates of the sampling sites, and bedding orientations of the sedimentary rocks were measured with a magnetic compass.

3. Laboratory Techniques

Rock magnetic experiments were performed in the Paleomagnetic Laboratory of Northwest University (Xi'an, China) and the Ivar Giæver Geomagnetic Laboratory (IGGL) at the University of Oslo (Norway). Isothermal remanent magnetization (IRM) acquisition experiments and thermal demagnetization of three-component IRMs (3-axis IRM; soft (<0.1 T), medium (0.1–0.4 T) and hard (0.4–2.5 T) coercivity fractions) were performed with an ASC IM-10-30 pulse magnetizer, JR-6A magnetometer and MMTD-48 furnace. k-T experiments were conducted in air using an MFK1 susceptibility meter and its CS-4 temperature control system. Anisotropy of magnetic susceptibility (AMS) measurements made using a MFK1 susceptibility meter. Scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) analyses were carried out at the State Key Laboratory of Continental Dynamics, Northwest University.

Thermal demagnetization experiments and alternating field (AF) demagnetization experiments were carried out at the Institute of Earth Environment, Chinese Academy of Sciences and the Paleomagnetic Laboratory of Northwest University. AF demagnetization were carried out on D-2000 alternating demagnetizer. After each demagnetization step, magnetic remanence measurements were made with a 2G-760 superconducting (cryogenic) magnetometer or a JR-6A spinner magnetometer. Vector analysis of the measured magnetic directions was performed using orthogonal vector diagrams (Zijderveld, 1967) and principal component analysis (PCA) (Kirschvink, 1980) to identify and quantify discrete magnetic components. Site mean paleomagnetic directions were calculated using standard Fisher statistics (Fisher, 1953) and analyzed via PMGSC (Enkin, 1994), PaleoMac 6.5 (Cogné, 2003) and Pmagpy (Tauxe et al., 2016) software.

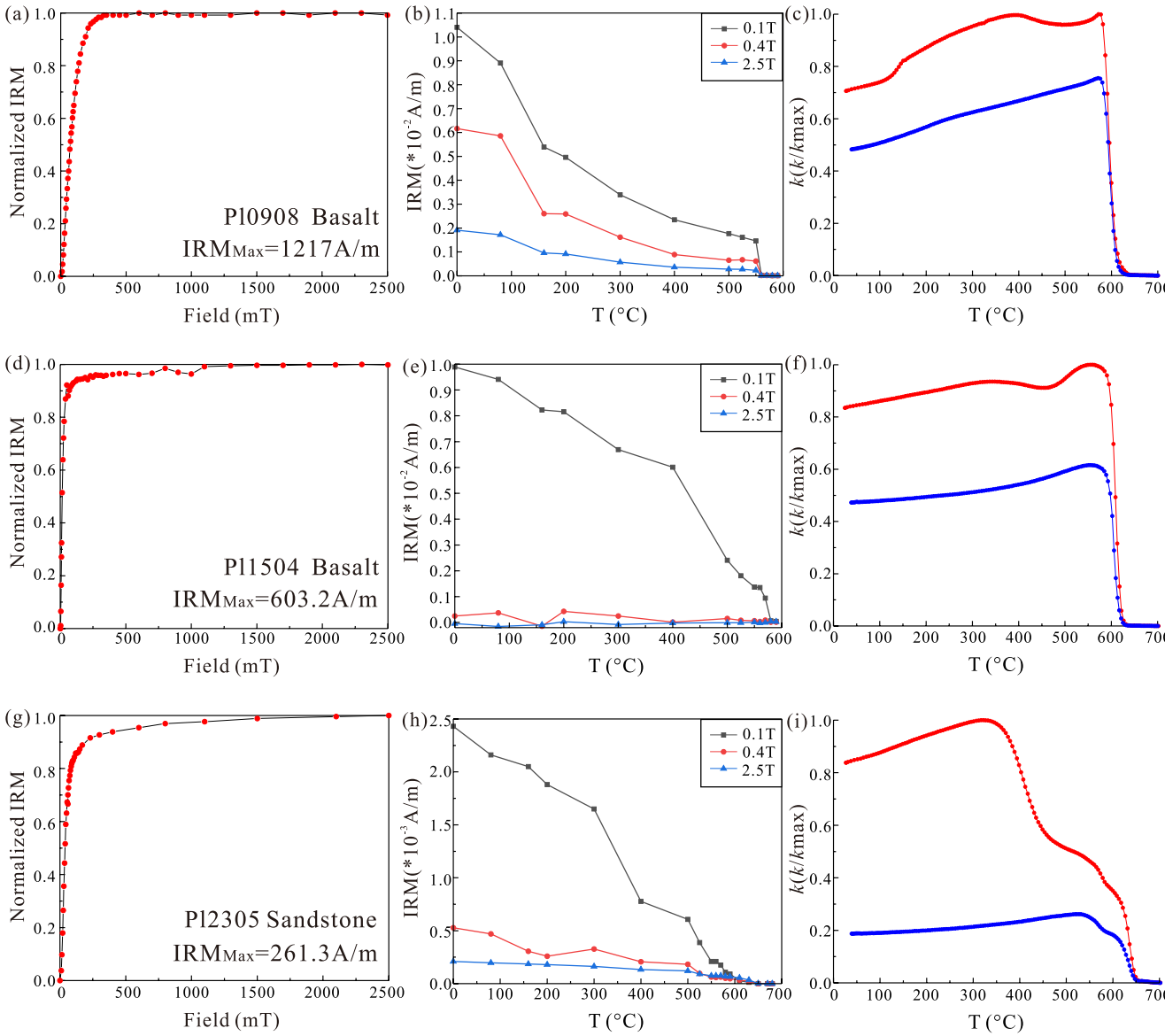


Figure 2. Isothermal remanent magnetization (IRM; a, d, g), thermal demagnetization of 3-axis IRM (b, e, h) and *k*-T (e, f, i) curves for representative rock samples. (a)–(f) basalt samples; (g)–(i) sandstone samples. Black, red and blue curves in the 3-axis IRM diagrams represent the soft, medium and hard coercivity fractions, respectively, while the red and blue *k*-T curves represent heating and cooling, respectively.

4. Results

4.1. Rock Magnetic Results

For the basalt samples, IRM acquisition curves rise steeply and reach saturation by 300 mT, indicating that they are dominated by a soft coercivity phase (Figures 2a and 2d). However, thermal demagnetization of three-component IRMs reveals that a sharp unblocking observed in the soft component at 580°C (Figures 2b and 2e) is also reflected in some of the medium and hard components, suggesting that magnetite is present in a range of coercivities (Figure 2b). Some of the soft- and medium-coercivity fractions also show a noticeable inflection at approximately 150°C (Figure 2b), which may be indicative of the presence of titanomagnetite. *k*-T curves show that the magnetic susceptibility of the basalt samples increases gradually with increasing temperature and exhibits a subtle Hopkinson peak before dropping rapidly at 570°C (Figures 2c and 2f). The shapes of the heating/cooling curves are similar, although the relative susceptibility is elevated after heating. SEM and EDS investigations of the basalts reveal that their constituent magnetic minerals mostly occur as iron oxides, but that

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some contain a small amount of titanium. These iron oxides are often euhedral to subrounded (Figures S1a–S1d in Supporting Information S1), indicating that they are likely primary and have not been strongly affected by chemical alteration and weathering.

For the tuffaceous sandstone samples, IRM acquisition curves rise steeply in low fields and reach over 85% of the saturation IRM by 200 mT, after which they continue to climb slowly but do not reach full saturation by 2.5 T (Figure 2g). Thermal demagnetization of three-component IRMs shows that the dominant, soft fraction fully demagnetizes by 580°C. *k*-T curves reveal that with increasing temperature, the magnetic susceptibility decreases abruptly in several intervals: at 350–400°C, 560–580°C and 580–650°C (Figure 2i). With cooling, the *k*-T curves are found to be irreversible, wherein the inflection at 300–450°C is absent. This likely reflects the breakdown of some metastable phase during heating; e.g., the inversion of maghemite to hematite. To consider if the remanent magnetic directions of the sandstones could have been biased by sedimentary compaction, we measured the AMS of 33 sandstone samples. They revealed low degrees of anisotropy ranging from 1.005 to 1.03, with a mean anisotropy degree (p') of 1.014 (Figures S1e and S1f in Supporting Information S1).

Rock magnetic results are not reported from the limestone samples because they did not yield interpretable paleomagnetic results (see next section).

4.2. Paleomagnetic

Seventy-nine limestone samples (8 sites) were subjected to demagnetization experiments. The intensity of their NRM ranged from $6.4 \times e^{-3}$ to $6.3 \times e^{-5}$ A/m and decreased rapidly with demagnetization. By 350°C, the residual intensity of the samples usually attenuated to less than 10% of the NRM. A low-temperature component (LTC) of remanent magnetization was found to be separable below 200°C, but above that temperature, the remanent directions became unstable, and characteristic remanent magnetization (ChRMs) could not be determined (Figure 3a).

The NRM intensity of 94 basalt samples (9 sites) ranged from $2.9 \times e^{1}$ to $4.3 \times e^{-4}$ A/m. Their remanent intensity decreased steadily during demagnetization and was completely lost by 570–590°C or 80 mT, consistent with the rock magnetic results and the interpretation that magnetite is the main magnetic carrier in these samples. Demagnetization revealed two stable and directionally distinct remanent components. A LTC (or a low-field component (LFC)), which was removed below 350°C (or 25 mT), is generally directed moderately downward to the north, whereas a high-temperature (high-field) ChRM, isolated between 350 and 590°C and 25–80 mT (Figures 3b–3d), is generally sub-horizontal and directed to the south-southwest (Table S1).

Sixty-eight sandstone samples (6 sites) were thermally demagnetized and their NRM intensity ranged from $5.9 \times e^{-1}$ to $8.2 \times e^{-2}$ A/m. The results of their demagnetization mostly revealed two component behavior. A LTC was removed by 350°C, and a high-temperature ChRM was isolated between 350°C and 575°C (Figures 3e and 3f). As in the basaltic samples, the LTC is oriented moderately down to the north, whereas the ChRM in these samples is sub-horizontal and directed toward the east. As in the basalts, the thermal demagnetization spectra of the ChRM in the sandstones suggests that it is carried by magnetite, and a significant contribution from hematite is not evident.

Twelve samples of conglomerate clasts (basalt) were thermally demagnetized and their NRM intensity ranged from $8.6 \times e^{-1}$ to $1.05 \times e^{-2}$ A/m. The results of their demagnetization mostly revealed two component behavior (Figures 3g and 3h), as seen in the other basalts. A LTC was removed by 350°C, and a high-temperature ChRM was isolated between 350°C and 590°C.

4.3. Directional Analysis

The LTC and LFC directions retrieved from all the sampled lithologies are similar, and considering them collectively at the sample level ($n = 208$), the mean direction is $D_g = 5.8°$, $I_g = 57.0°$, $k_g = 12.8$, $\alpha_{95g} = 2.8°$ in geographic coordinates, and $D_s = 33.4°$, $I_s = 17.0°$, $k_s = 11.3$, $\alpha_{95s} = 3.0°$ in tilt-corrected coordinates. In geographic coordinates, this mean LTC and LFC direction is close to the direction of the present-day dipole field ($D_{PDF} = 0°$, $I_{PDF} = 52.6°$; Figures 4a and 4b) and we interpret it as a recent viscous magnetic overprint.

The ChRMs of the basalt and sandstone are dissimilar. The mean ChRM direction from the 9 basalt sites is $D_g = 194.6°$, $I_g = 9.9°$, $k_g = 59.9$, $\alpha_{95g} = 6.7°$ in geographic coordinates, and $D_s = 172.8°$, $I_s = 39.1°$, $k_s = 80.1$,

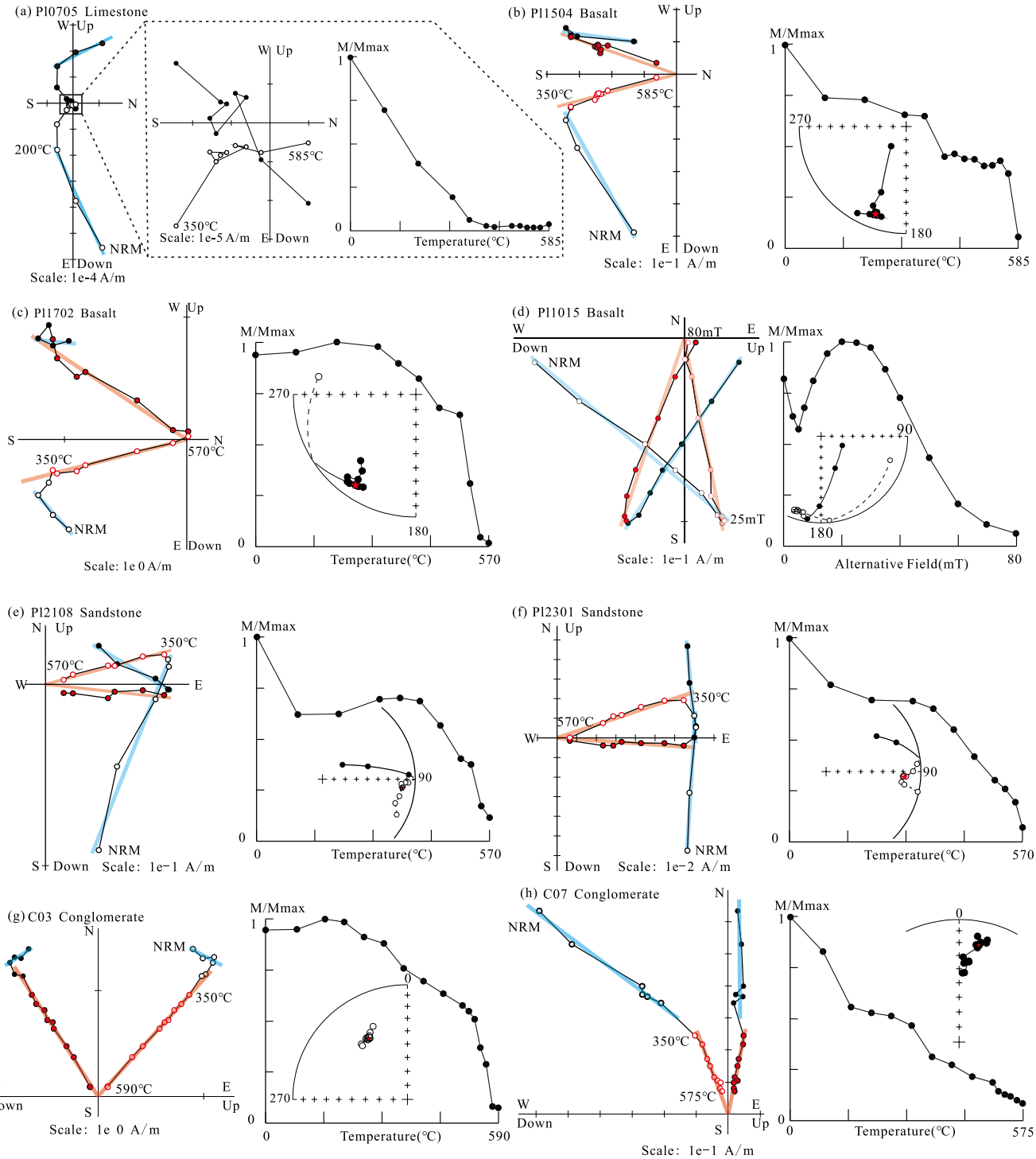


Figure 3. Representative thermal demagnetization results of samples from the Lugu Formation (shown in geographic coordinates). Filled circles represent the horizontal component in the orthogonal vector diagram and lower hemisphere projections for the stereonets, while open circles represent the vertical components/upper hemisphere projections. The colored circles and colored straight lines represent the points involved in the PCA and the fitted direction, respectively. Blue (red) corresponds to the LTCs/LFC (ChRMs).

$\alpha_{95s} = 5.8°$ after tilt correction (Figures 4c and 4d, Table 1). The mean ChRM direction from the 6 sandstone sites is $D_g = 84.9°$, $I_g = -5.8°$, $k_g = 51.1$, $\alpha_{95g} = 9.5°$ in geographic coordinates, and $D_s = 91.3°$, $I_s = -47.9°$, $k_s = 61.3$, $\alpha_{95s} = 8.6°$ after tilt correction (Table 1). Alternatively computing a mean ChRM direction from the $n = 40$ sandstone samples (rather than sites, since individual samples may partly average secular variation) yields:

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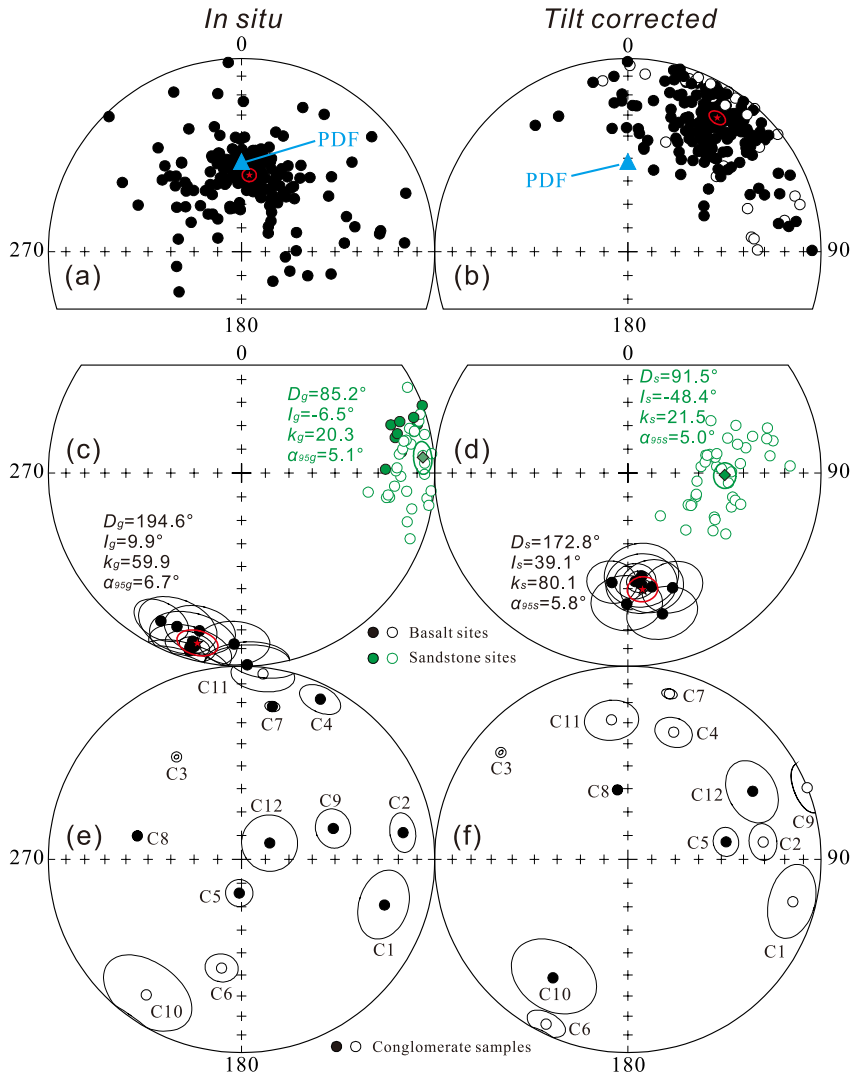


Figure 4. (a) and (b) Equal-area plots of the low stability components (LSC) (from the limestone, basalts, sandstones and conglomerate cobbles) before and after tilt correction; red star is the mean direction of the LSC; blue triangle is the mean direction of the local present-day time-averaged dipole field (PDF). (c) and (d) Equal-area stereographic projection of the basalt site means (black) and sandstone sample (green) ChRMs before and after tilt correction; red star and green rhombus represent the mean ChRM directions of the basalt/sandstone populations, respectively. Open (closed) circles are projections on the upper (lower) hemisphere. (e) and (f) Equal-area projection of the conglomerate clast ChRM directions before and after tilt correction.

$D_g = 85.2°$, $I_g = -6.5°$, $k_g = 20.3$, $\alpha_{95g} = 5.1°$ in geographic coordinates and $D_s = 91.5°$, $I_s = -48.4°$, $k_s = 21.5$, $\alpha_{95s} = 5.0°$ after tilt correction (Figures 4c and 4d, Table 1).

In both geographic and tilt-corrected coordinate systems the basalt and sandstone mean directions differ by ~80–110° in declination, whereas the absolute values of their inclinations are more similar. We test the significance of these similarities by rotating each of these populations about a vertical axis until their mean declination equals zero and then conduct tests for a common mean direction. Using the site-level populations we find that their inclinations are statistically indistinguishable in both coordinate systems (although only marginally so in tilt-corrected coordinates), whereas with use of the sample-level sandstone data we find the mean inclination of the sandstone and basalt to be distinct at the 95% confidence level (Figure S2 in Supporting Information S1). Application of an inclination-only foldtest (Enkin & Watson, 1996) yielded inconclusive results, which is unsurprising given the similar structural attitude of all the sampled units.

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Table 1
ChRM Directions From the Lugu Formation Basalt and Sandstone Samples (Reference Point: 33.2°N, 86.9°E)

		Bedding			Paleomagnetic direction of ChRM							
Site ID	Lithology	Strike	dip	*n*/*N*	D_g	I_g	D_s	I_s	k	α_{95}	λp	φp
Pl09	Basalt	326	53	9/9	195.2	7.7	174	42.8	113.3	4.9	−31.7	93.3
Pl10	Basalt	326	53	11/16	182.5	12.4	158.7	36.4	16.2	11.7	−32.8	110.8
Pl11	Basalt	326	53	10/12	196.2	10.2	172.4	45.3	12.0	14.5	−29.6	94.7
Pl12	Basalt	326	53	9/11	196.5	6.8	176.2	43.1	41.6	8.1	−31.6	90.9
Pl13	Basalt	326	53	3/6	178.3	0.8	166.2	25.9	93.5	12.8	−41.3	104.9
Pl14	Basalt	337	51	6/8	196.3	4.6	180.5	32.7	26.7	13.2	−39.0	86.3
Pl15	Basalt	337	51	10/11	194.9	16.6	168.1	39.6	16.6	12.2	−33.2	100.1
Pl16	Basalt	342.5	56	7/12	202.8	14.9	173	41.5	20.4	13.7	−32.5	94.5
Pl17	Basalt	341.5	45	8/9	208.6	13.8	188.3	42.2	33.4	9.7	−31.7	77.3
Pl18	Sandstone	351	40	7/8	85.1	−17.6	88.2	−57.5	84.2	6.6	−18.5	210.9
Pl19	Sandstone	351	40	8/9	89.1	−8.5	93	−48	23.5	11.7	−18.5	200.9
Pl20	Sandstone	332	49	6/16	79.7	−5	90.6	−50.6	19.1	15.7	−17.0	203.6
Pl21	Sandstone	332	45	6/9	83.0	−1.9	91.6	−43.0	28.6	12.7	−14.6	197.5
Pl22	Sandstone	342.5	45	6/16	85.5	13.0	87.3	−30.8	25.9	13.4	−6.8	192.3
Pl23	Sandstone	342.5	45	7/10	87.2	−14.6	99.3	−57.1	19.1	14.2	−26.2	206.4
Basalt sites mean				9/9	194.6	9.9			59.9	6.7		
							172.8	39.1	80.1	5.8		
Mean of VGPs				9/9						5.8	−34.1	94.7
Sandstone sites mean				6/6	84.9	−5.8			51.1	9.5		
							91.3	−47.9	61.3	8.6	−17.0	201.8
Sandstone samples mean				40/68	85.2	−6.5			20.3	5.1		
							91.5	−48.4	21.5	5.0	−18.0	202.1

Note. *n*/*N* is the number of samples/sites used in the mean calculation and total number of samples/sites; D_g, I_g, D_s, and I_s, are the declinations and inclinations before and after tilt correction, respectively; k and α_{95}, the best estimate of the precision parameter and the radius of the 95% cone of confidence about the estimated mean direction, respectively; λp and φp are the latitude and longitude of VGP.

Converting the tilt-corrected ChRMs from the basalt and sandstones to virtual geomagnetic poles (VGPs), we compute a mean paleomagnetic pole for the basalt sites: $\lambda = -34.1°$N, $\varphi = 94.7°$E, $A_{95} = 5.8°$ and for the sandstone samples: $\lambda = -18.0°$N, $\varphi = 202.1°$E, $A_{95} = 5.6°$ (Table 1). With respect to the theoretical n-dependent A_{95} bounds proposed by Deenen et al. (2011), both of these poles appear consistent with the expectations of a pole that has sufficiently averaged paleosecular variation (PSV); the A_{95} min-max bounds are 5°–21° and 3°–8° for $N = 9$ and $n = 40$, respectively (the sandstone paleomagnetic pole computed from site-level data yields an A_{95} of 7.4° which is also falls within the min-max bounds for $N = 6$).

The 12 ChRM directions determined from conglomerate cobbles appear scattered (Figures 4e and 4f). A random distribution test reveals their resultant vector length ($R = 4.09$) to be less than the critical R-value of 6.55 ($n = 12$), confirming that they can be considered random at the 99% confidence level (Watson, 1956). This constitutes a positive conglomerate test, suggesting that the basaltic cobbles have retained a magnetization pre-dating sedimentation of the host conglomerate.

5. Discussion

5.1. The Nature and Age of the ChRMs

Because the ChRMs of the basalts and sandstones significantly differ (in both declination and inclination, and in both geographic and tilt-corrected coordinates), it is unlikely they represent magnetizations of the same origin

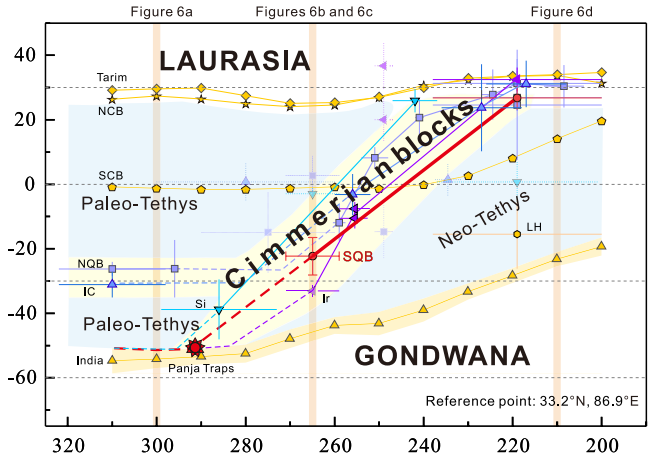


Figure 5. Paleolatitude evolution of the SQB and its surrounding blocks during the early Permian to Late Triassic, from the perspective of a common reference point, see Table S3 for detailed paleomagnetic data. Symbols: rhombus = Tarim; five-pointed star = North China Block; pentagon = South China Block; square = North Qiangtang Block; blue triangle = Indochina; inverted triangle = Sibumasu; red circle = South Qiangtang Block; oblique triangle = Iran; hexagon = Lhasa; black triangle = India. The large, nine-pointed red star highlights the inferred position of South Qiangtang along the northern margin of Gondwana (prior to its rifting).

and age; we therefore consider them individually. Although we have no field-tests that can directly constrain the age of the basalt ChRM, the positive conglomerate test on basaltic cobbles reveals that this lithology could have escaped remagnetization, and thus the basalt HTC could be primary. We note that the basalt ChRM directions are exclusively of reversed polarity, which is consistent with them having been acquired during the Kiaman Reversed Polarity Superchron (KRS), which ended at ~267-265 Ma (Belica et al., 2017; Hounslow & Balabanov, 2018). A primary to pre-Late Triassic age for the basalt ChRM is also supported by the observation that its mean direction (in both geographic and tilt-corrected coordinates) differs from all existing (Late Triassic and younger) paleomagnetic records from the SQB (e.g., Chen et al., 2017; Cao et al., 2019, 2020; Meng et al., 2018; Song et al., 2012; Table S2). Specifically, the basalt ChRM corresponds to a southern paleolatitude in either coordinate system (5.0°S in geographic coordinates; 22.3°S after tilt-correction), whereas the Late Triassic and younger paleomagnetic data from the SQB exclusively place that block north of 25°N. Although we cannot presently rule out the possibility of a late Permian to mid-Triassic remagnetization, given the positive conglomerate test and the exclusively reversed polarity of the basalts, we favor the interpretation that their magnetization is primary (middle Permian). These paleomagnetic data (Table 1 and Figure 5) provide a position at a latitude of 22.3 ± 5.8°S in the Southern Hemisphere for the SQB in the middle Permian.

The nature of the tuffaceous sandstone ChRM is more ambiguous. Note that the positive conglomerate test—although discounting a regional thermal remagnetization—does not exclude a possible chemical remagnetization of the sandstones, because the sampled conglomerate clasts are of a different lithology (basalt). If we first assume that the ChRM could be primary and adopt the tilt-corrected directions, we observe that their inclinations are exclusively negative, suggesting that they were not acquired during the KRS. Although the directions could have been acquired immediately after the KRS, this was an interval of mixed field polarity (Hounslow & Balabanov, 2018; Shen et al., 2019). If both the basalt and sandstone ChRMs are primary, their declination differences would have to be due to local vertical axis rotations, as an overall rotation of the entire SQB would otherwise need to be exceedingly rapid (>10°/Myr). Such local vertical-axis rotations must have occurred in a relative sense within the PL section (i.e., between the basalt and sandstone outcrops sampled), but we did not observe any such accommodation structures in the field. Furthermore, the orientation of the basalt and sandstone units are similar (both dipping to the northeast), indicating that any strong relative rotations must have occurred prior to tilting. A final indirect argument against a primary origin for the sandstone ChRM is that the tilt-corrected inclinations of the stratigraphically-younger sandstones are steeper than those of the basalts, and thus would place the SQB at higher southern latitudes (28.4°S). This conflicts with the geologic evidence suggesting that the SQB was drifting northward with time (see next section).

If the sandstone ChRM represents a chemical remagnetization, it is possible that it was acquired during or after tilting, making its inclination comparable to or shallower than that of the basalts. Interestingly, the mean inclination of the ChRM in geographic coordinates corresponds to an equatorial latitude (~3°N or S). According to a simple interpolation between the paleolatitude determined from the basalts and a Late Triassic paleolatitude estimate for the SQB by Song et al. (2012), the SQB was in equatorial latitudes in the Early to Middle Triassic (Figure 5, see next section). One possibility is thus that the sandstones were chemically remagnetized during pre-Late Triassic tectonism, in which case the declination difference between the basalts and sandstones could be explained by a longer-term rotation of the entire SQB (at rates of ~2–4°/Myr). However, owing to the uncertainties in the nature and age of the sandstone ChRM, it cannot be used for paleogeographic reconstruction. Further investigations of the origin of this magnetization present an opportunity for future work. Because we cannot exclude the possibility of vertical axis rotations having affected the sampling area, we also refrain from using the declination of the basalt ChRM for paleogeographic reconstruction, and instead only use its inclination to constrain the paleolatitude of the SQB.

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5.2. Paleogeographic Implications

With our middle Permian paleomagnetic result from the Lugu Fm. basalts providing the first quantitative paleolatitude estimate of the SQB in pre-Late Triassic time, we now turn to consider its paleogeographic implications by placing it in the context of existing paleomagnetic and geologic constraints from the Tethyan realm. To achieve this, we compiled late Carboniferous to Triassic paleomagnetic data from the major cratons and blocks that may have surrounded the SQB in late Paleozoic-early Mesozoic time (Table S3), and computed their corresponding paleolatitudes relative to a common reference point at our PL section (33.2°N, 86.9°E; Figure 5). Using these data together with summarized geological observations, we consider the position of the SQB and its paleogeographic implications at three key points in time: the beginning of the Permian (300 Ma), the middle Permian (265 Ma) and the Late Triassic (210 Ma).

At the start of the Permian (~300 Ma), the Tarim Block (Wei et al., 2020) and North China Block (NCB) were located at low to mid-latitudes in the Northern Hemisphere, close to the southern margin of Laurasia, the South China Block (SCB) was at the equator, the NQB and Indochina (IC) were positioned at low southern latitudes, and India was located at mid-latitudes of the Southern Hemisphere (Figure 5). Similarities in the paleolatitudes and geologic records of the SCB, IC and NQB suggest they may together have formed an archipelago at this time, but the absolute longitude of these blocks remains unknown. In Figure 6a we show a conventional 'cylindrical' reconstruction that keeps these blocks close to the longitude of the NCB, following Huang et al. (2018).

Although there are no quantitative constraints on the paleolatitude of the SQB at this time, its independence from the NQB is inferred from their differing geological records. The late Carboniferous-early Permian successions of the SQB include glacial relics (Li et al., 1995; Liang et al., 1982; Zhang, Wang, et al., 2012) and bear cold water organisms, including peri-Gondwanan fusulinids (Zhang, Wang, et al., 2013). In contrast, the Carboniferous-Permian sedimentary succession of the NQB are characterized by stable platform-type carbonate deposits that include warm-water faunas (Zhang et al., 2016). The Carboniferous-Permian glacial deposits of the SQB and its peri-Gondwanan faunas point to an origination of the SQB from the northern margin of Gondwana. This interpretation is further substantiated by recent work that has determined the SQB to be floored by a Pan-African ('Gondwanan') crystalline basement (Dong et al., 2011; Li, 2003; Liu et al., 2016; Pullen et al., 2011). Records of late Pan-African magmatic activity have also been recognized in the SQB and may be linked to tectonism that was then occurring along the northern margin of Gondwana, notably along the margin of India (Cawood et al., 2007; Liu, 2017). In addition to the fossil and glacial evidence, the results of sedimentary provenance studies also suggest that the SQB remained connected with Gondwana until the early Permian (Fan et al., 2014, 2015; Zhang, Wang, et al., 2013). Accordingly, we reconstruct the SQB to the northern edge of Gondwana during the late Carboniferous-early Permian (Figure 6a).

The latitudinal separation between the NQB and the southern margin of Tarim/Laurasia in Figure 6a indicates that a large 'Paleo-Tethys Ocean' laid between them then, the remnants of which may now be preserved along the Xijinwulan-Jinshajiang Suture Zone (XJS) that demarcates the northern margin of the NQB in the Qinhai-Tibet Plateau. However, the separation between the NQB and SQB reveals that the southern branch of the Paleo-Tethys ('Longmu Co-Shuanghu Ocean') was also an important basin at this time.

The spalling of the SQB from the northern margin of Gondwana may have begun in the early Permian, when a swarm of mafic dikes (thought to relate to a mantle plume and termed the Qiangtang-Panjal large igneous province) intruded the SQB (Dan et al., 2021; Wang et al., 2014, 2019; Xu et al., 2016; Zhai et al., 2013), ultimately weakening the crust and allowing the block to be detached and pulled northward by slab pull (Wan et al., 2019, 2021). This may have occurred in concert with a string of other marginal blocks (together forming the Cimmerian continent) that rifted from the northern margin of Gondwana in the early Permian. Although simple, this model is consistent with some of the sparse paleomagnetic data assembled from those blocks (Figure 5, Table S3).

Shen et al. (2016) reported Cathaysian affinity warm-water brachiopod faunas in limestones of the Lugu Formation, indicating that the SQB had drifted to temperate latitudes by the upper Kungurian/lower Guadalupian. This is consistent with our new middle Permian paleomagnetic result from the Lugu Formation basalts, which place the SQB at 22.3 ± 5.8°S at this time (Figure 6b). Similar middle Permian faunas have also been reported from Iran, Pamir and other continental blocks associated with the Cimmerian continent, or what Angiolini et al. (2013) term the 'Trans-Himalayan Province'. The appearance of these faunas of Cathaysian affinity also suggests that the

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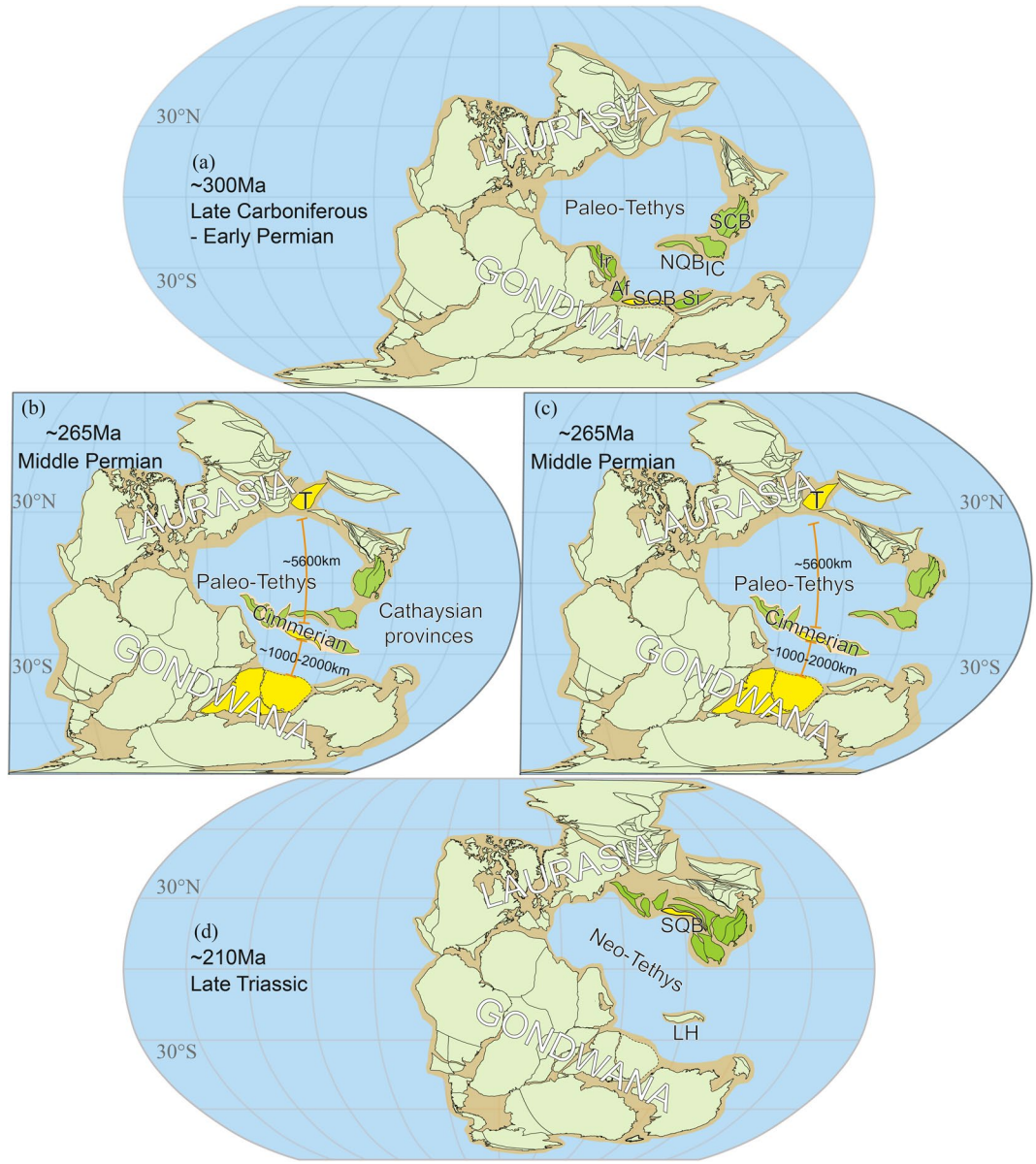


Figure 6. Paleogeographic reconstructions of the South Qiangtang Block (SQB) and its surrounding blocks during the late Carboniferous-early Permian to Late Triassic. In all panels, the position of the SQB is positioned according to the data presented in the main text. In panel c) the positions of the South China Block (SCB), Indochina (IC) and North Qiangtang Block (NQB) are modified after Zhang and Torsvik (2022). All other blocks in all other panels are modified after Huang et al. (2018). Other abbreviations: Ir = Iran; Af = Afghanistan; Si = Sibumasu; H = Lhasa; T = Tarim.

SQB was approaching the SCB, which the NQB likely remained proximal to (Shen et al., 2016). Support for the latter interpretation can be found in late Permian strata of the NQB, which bear typical genera of the Cathaysian flora associated with the SCB, such as *Gigantopteris*, *Pecopteris*, *Gigantonoclea*, *Rhizomopsis*, *Sphenophyllum*, *Palaeofusulina* and *Colaniella* (Li & Yao, 1980; Qiao et al., 2021; Wen, 1979; Zhang, Wang, et al., 2013).

The northward drift of the Cimmerian continent marked the opening and spreading of the Neo-Tethys Ocean (between the Cimmerian continent and northern Gondwana) at the expense of a shrinking Paleo-Tethys Ocean (between the Cimmerian continent and southern Laurasia). According to our new paleomagnetic result, the northward drift of the SQB allowed the Neo-Tethys Ocean to reach a width of ~1,000–2,000 km by middle Permian time (a more precise width is difficult to estimate given the substantial uncertainties on the original size of the northern margin of Greater India and the Himalayas) (Meng et al., 2019, 2020). At the same time, given a

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conventional 'cylindrical' longitudinal reconstruction of the SCB-IC-NQB (Huang et al., 2018), this positioning of the SQB would imply that the Longmu Co-Shuanghu branch of the Paleo-Tethys had nearly disappeared by this point, whereas the Jinshajiang branch had a width of >5,500 km (Figure 6b). Alternatively, in Figure 6c we show a contrasting reconstruction wherein the SCB-IC-NQB archipelago was located further east in earliest Permian time (Zhang & Torsvik, 2022). In this scenario, the Longmu Co-Shuanghu ocean could have remained open and significant in width at this time (Figure 6c).

During late Permian to Late Triassic time, the NQB and SQB moved swiftly northward, leading to a rapid expansion of the Neo-Tethys Ocean and a progressive shrinkage of the Paleo-Tethys Ocean (Zhou et al., 2019). More specifically, using our new result and the next oldest paleomagnetic constraint from the SQB (Late Triassic), we calculate that the SQB moved northward at an average rate of ∼11.7 cm/yr between 265 and 219 Ma. Late Triassic paleomagnetic data from the SQB, NQB and IC yield indistinguishable paleolatitudes, suggesting that they had merged into a unified whole by then (Song et al., 2012, 2015; Yan et al., 2019), as has also been inferred geologically (Ding et al., 2013; Zhao et al., 2018) (Figure 6d). The Triassic collision of the SQB and NQB marked the final closure of the Longmu Co-Shuanghu Ocean, now demarcated by the LSS, whereas the probably broadly coeval collision between the NQB and greater Tarim-NCB marked the closure of the Jinshajiang Ocean, now preserved along the XJS.

6. Conclusions

We presented new paleomagnetic results from middle Permian volcanics and sedimentary rocks of the SQB. Although the origin of the magnetization of the sedimentary rocks is ambiguous, a conglomerate test and the polarity of the basalts indicates they likely preserve a primary magnetization acquired during the Kiaman Reversed Polarity Superchron. Using these data, we provide the first quantitative paleolatitude constraint on the SQB in pre-Mesozoic time: estimating that it was located at 22.3 ± 5.8°S at 265 Ma. With reference to the larger paleogeographic context at this time, this position of the SQB indicates that the Neo-Tethys Ocean had a width of ∼1,000–2,000 km in the middle Permian, while the Paleo-Tethys Ocean spanned ∼5,500 km. However, a determination of the relative importance of the northern and southern branches of the Paleo-Tethys Ocean are sensitive to the longitudinal position of the NQB, which probably formed an archipelago together with the SCB and IC. Following its rifting from Gondwana in the early Permian, the SQB, together with the wider Cimmerian continent, drifted rapidly northwards until the Late Triassic, by which time a series of block collisions, including the merger of the SQB and NQB, had led to closure of the various branches of the Paleo-Tethys Ocean.

Data Availability Statement

Data used in this study (Tables S1–S3 in Supporting Information) can be accessed at https://doi.org/10.5281/zenodo.7013623).

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